

Mathematical Analysis — Lecture 1-4

Class 5% Homework 15% Quizzes 20% Midterm 20% Final 40%

Preliminary

- $\mathbb{N}$ denotes the set of all positive integers.
- $\mathbb{Z}$ denotes the set of all integers.
- $\mathbb{Q}$ denotes the set of all rational numbers.
- $\mathbb{R}$ denotes the set of all real numbers.

Limits and Continuity

1. Key idea

We use limits to describe the tendency of a function as the variable approaches a given point, but not the value at the point.

2. Important examples

Example 1: Same limit, different value at the point

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

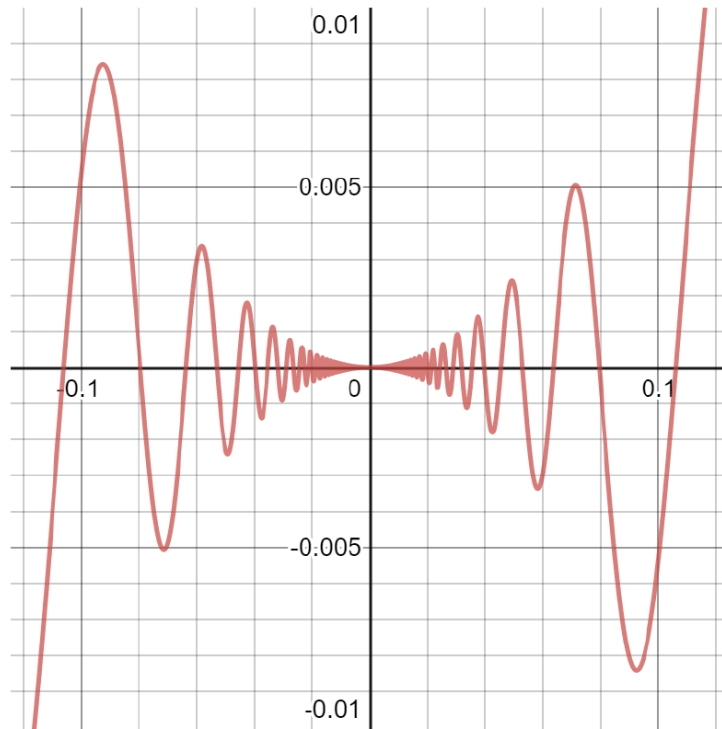


Figure: $f(x) = x^2 \sin \frac{1}{x}$ Zoom in

$$g(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

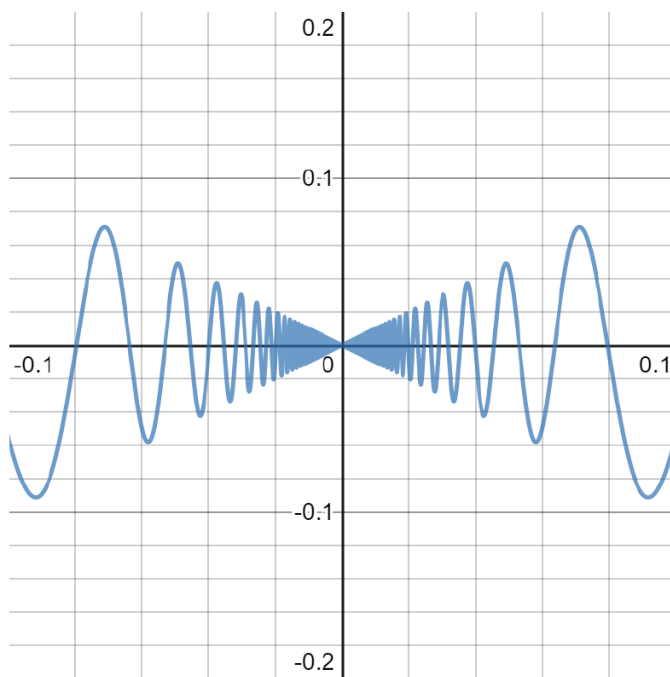


Figure: $g(x) = x \sin \frac{1}{x}$ Zoom in

Useful estimates:

$$|f(x)| \leq |x|^2, \quad |g(x)| \leq |x|.$$

Therefore:

$$\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 0} g(x) = \underline{\hspace{2cm}}.$$

My takeaway:

If x gets close to 0, keeping $x \neq 0$, $f(x)$ approaches 0 with error $\leq |x|^2$; $g(x)$ approaches 0 with error $\leq |x|$.

Example 2: Average speed and instantaneous speed

Average speed:

$$\frac{\Delta s}{\Delta t} = \frac{f(t_2) - f(t_1)}{t_2 - t_1}.$$

With $h = t_2 - t_1$:

$$\frac{f(t_1 + h) - f(t_1)}{h}.$$

Instantaneous speed at t_0 :

$$v(t_0) = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h} = \lim_{t \rightarrow t_0} \frac{f(t) - f(t_0)}{t - t_0}.$$

Informal definition of limit

Assume $f(x)$ is defined on an open interval about c , except possibly at c itself. If $f(x)$ is arbitrarily close to L for all x sufficiently close to c other than c itself, then

$$\lim_{x \rightarrow c} f(x) = L.$$

This is read “the limit of $f(x)$ as x approaches c is L .”

3. Limit calculation techniques

Algebraic simplification

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = \underline{\hspace{2cm}}.$$

Rationalizing the denominator

$$\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x} + 1} = \underline{\hspace{2cm}}.$$

Remark: Simplification when calculating limits.

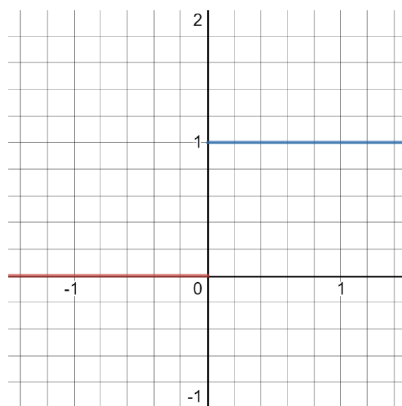
General strategy:

1. Check direct substitution.
2. Factor or rationalize if necessary.
3. Simplify, then evaluate the limit.

Class 2: Describing the behavior of a function as $x \rightarrow 0$

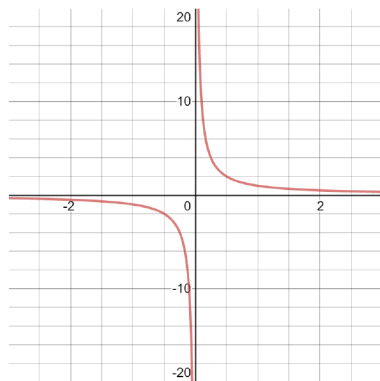
1. Jump

$$\text{Unit step function } U(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$



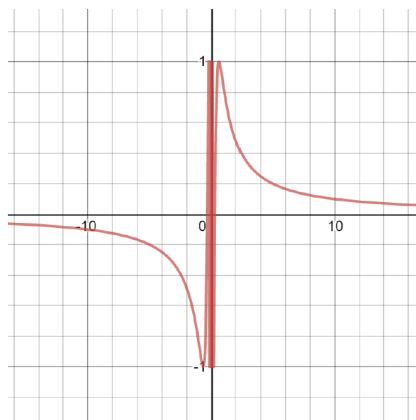
2. Unbounded

Unbounded function $g(x) = \begin{cases} \frac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases}$



3. Oscillation

Oscillating function $h(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases}$



4. When a limit does not exist

Type	What happens near the point?	Example
Jump	Left and right sides approach different values	Unit step function
Unbounded	Function values grow without bound	$1/x$ near 0
Oscillation	Values do not approach one fixed number	$\sin(1/x)$ near 0

A few words

I will take notes in class every time, and I will post them to my personal website.
The notes will include the main points of the lecture, and I will try to make

them as clear as possible. However, I will not be able to include all the details, so please make sure to attend the lectures. Also, I will not be writing down every single example, so please make sure to read the textbook and do the exercises.

Lecture 2

Definition of limit (Informal)

$$g(x) = \begin{cases} 2x, & \text{if } x \neq 4, \\ 0, & \text{if } x = 4. \end{cases}$$

By the informal description of a limit: if x gets sufficiently close to 4, but is not equal to 4, then $g(x)$ gets arbitrarily close to 8.

We need to show that the gap between $g(x)$ and $L = 8$, namely $|g(x) - 8|$, can be made as small as we choose if x is kept close enough to $c = 4$, but is not equal to 4.

Definition

Let $f(x)$ be defined on an open interval I about c , except possibly at c itself. We say that the limit of $f(x)$ as x approaches c is the number L , and write

$$\lim_{x \rightarrow c} f(x) = L,$$

if, for every $\varepsilon > 0$, there exists a corresponding $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever } x \in I \text{ and } 0 < |x - c| < \delta.$$

Simpler writing

We say $\lim_{x \rightarrow c} f(x) = L$ if, for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\text{if } x \in I, \quad 0 < |x - c| < \delta, \quad \text{then } |f(x) - L| < \varepsilon.$$

Personal interpretation

- ε : how close we require $f(x)$ to be to L .
- δ : how close x needs to be to c in order to satisfy that requirement.
- $|f(x) - L| < \varepsilon$: $f(x)$ lies within an ε -neighborhood of L .
- $0 < |x - c| < \delta$: x lies within a δ -neighborhood of c , but $x \neq c$.

The key idea is

Given any $\varepsilon > 0$, we can find a $\delta > 0$
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such that

$$x \text{ is sufficiently close to } c \implies f(x) \text{ is within } \varepsilon \text{ of } L.$$

In other words, ε is the required accuracy for the output, while δ is how much accuracy we need for the input.

Thus, the ε - δ definition gives a precise mathematical meaning to the intuitive statement: if x is sufficiently close to c , then $f(x)$ is sufficiently close to L .

Useful remarks

We shall often use the following arguments:

- (i) Let a be a real number. If, for every $\varepsilon > 0$, we have $a \leq \varepsilon$, then $a \leq 0$.
- (ii) Let a be a real number. If, for every $\varepsilon > 0$, we have

$$-\varepsilon \leq a \leq \varepsilon,$$

then $a = 0$.

Example: revisiting $g(x)$

Consider

$$g(x) = \begin{cases} 2x, & \text{if } x \neq 4, \\ 0, & \text{if } x = 4. \end{cases}$$

Show that

$$\lim_{x \rightarrow 4} g(x) = 8.$$

Analysis. For every $\varepsilon > 0$, we need

$$|g(x) - 8| = |2x - 8| < \varepsilon, \quad x \neq 4,$$

which requires

$$0 < |x - 4| < \frac{\varepsilon}{2}.$$

Proof. Given arbitrary $\varepsilon > 0$, take

$$\delta = \frac{\varepsilon}{2}.$$

Then

$$|g(x) - 8| = |2x - 8| = 2|x - 4| < 2\delta = \varepsilon$$

whenever $0 < |x - 4| < \delta$. By the definition of limit,

$$\lim_{x \rightarrow 4} g(x) = 8.$$

Exercise

Prove that

$$\lim_{x \rightarrow 1} \frac{1}{x} = 1.$$

Hint. Restrict x by taking $x > \frac{1}{2}$. Then

$$\left| \frac{1}{x} - 1 \right| = \frac{|x - 1|}{|x|} < 2|x - 1|.$$

Exercise 1.1.5 revisited

Using the definition of limit, prove that the following functions have no limit as $x \rightarrow 0$.

(a)

$$U(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

(b)

$$g(x) = \begin{cases} \frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

(c)

$$f(x) = \begin{cases} 0, & x \leq 0, \\ \sin \frac{1}{x}, & x > 0. \end{cases}$$

Tip. Suppose the limit exists and derive a contradiction.

For (b)

Suppose

$$\lim_{x \rightarrow 0} \frac{1}{x} = L.$$

Use the definition of the limit with $\varepsilon_0 = 1$.

Lecture 3

Date: 2026.9.10

Theorem 1.1.3 (Limit Laws)

Assume that

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M.$$

(a) **Sum Rule**

$$\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x).$$

Proof. Let $\varepsilon > 0$. Choose $\delta_1, \delta_2 > 0$ so that $|f(x) - L| < \varepsilon/2$ and $|g(x) - M| < \varepsilon/2$. For $\delta = \min\{\delta_1, \delta_2\}$,

$$|(f(x) + g(x)) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \varepsilon.$$

(b) Product Rule

$$\lim_{x \rightarrow c} (f(x) \cdot g(x)) = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x).$$

Proof. Since $f(x) \rightarrow L$, it is bounded near c ; take a neighborhood where $|f(x)| \leq |L| + 1$. Then

$$|f(x)g(x) - LM| \leq |f(x)||g(x) - M| + |M||f(x) - L|,$$

and choosing the two errors sufficiently small makes the right side less than any prescribed ε .

(c) Quotient Rule

If $M \neq 0$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}.$$

Proof. Since $g(x) \rightarrow M \neq 0$, eventually $|g(x)| > |M|/2$. Hence

$$\left| \frac{1}{g(x)} - \frac{1}{M} \right| = \frac{|g(x) - M|}{|g(x)||M|} \leq \frac{2|g(x) - M|}{|M|^2} \rightarrow 0.$$

Thus $1/g(x) \rightarrow 1/M$, and the quotient rule follows from the product rule.

Remark: Does the Converse Hold?

The Sum Rule requires that both limits on the right-hand side exist. In general, the converse is false: the existence of

$$\lim_{x \rightarrow c} (f(x) + g(x))$$

does not imply that both $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exist.

Counterexample

Let

$$f(x) = \sin \frac{1}{x}, \quad g(x) = -\sin \frac{1}{x}.$$

Consider $x \rightarrow 0$. Although

$$\lim_{x \rightarrow 0} \sin \frac{1}{x}$$

does not exist, we have

$$f(x) + g(x) = 0.$$

Therefore,

$$\lim_{x \rightarrow 0} (f(x) + g(x)) = 0$$

exists. Thus,

$$\boxed{\lim(f + g) \text{ exists} \not\Rightarrow \lim f, \lim g \text{ both exist}}$$

The problematic behavior of two functions can cancel each other out.

Exercise 1.1.9 (Limits of Polynomials)

If

$$P(x) = \sum_{k=0}^n a_k x^k = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

then

$$\lim_{x \rightarrow c} P(x) = \sum_{k=0}^n a_k c^k = P(c).$$

Exercise 1.1.10 (Limits of Rational Functions)

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}.$$

Theorem 1.1.4 (The Sandwich Theorem)

Suppose that

$$g(x) \leq f(x) \leq h(x)$$

for all x in some open interval containing c , except possibly at $x = c$. Suppose also that

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then

$$\lim_{x \rightarrow c} f(x) = L.$$

Proof. Given $\varepsilon > 0$, choose $\delta > 0$ so that $L - \varepsilon < g(x)$ and $h(x) < L + \varepsilon$ whenever $0 < |x - c| < \delta$. The inequalities $g(x) \leq f(x) \leq h(x)$ then imply $|f(x) - L| < \varepsilon$.

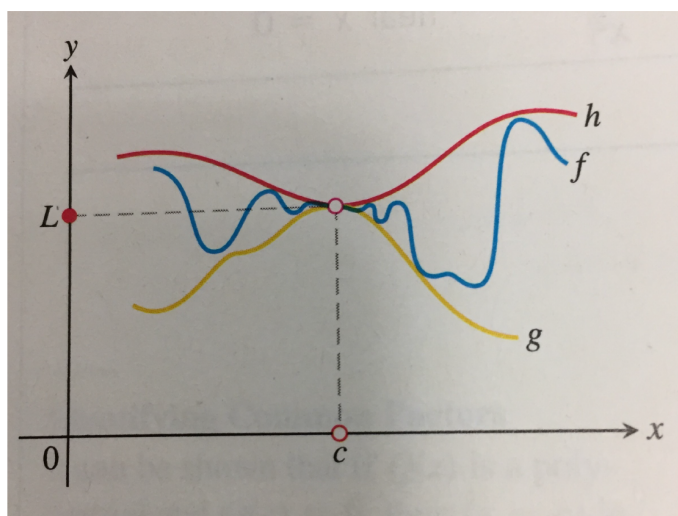


Figure: The Sandwich Theorem: $g(x) \leq f(x) \leq h(x)$

Exercise 1.1.12

Assume I is an open interval, $c \in I$, and

$$f(x) \leq b \quad \text{for all } x \in I \setminus \{c\}.$$

Then

$$\lim_{x \rightarrow c} f(x) \leq b,$$

provided the limit exists.

Exercise 1.1.13 (Order-preserving Property of Limits)

If

$$f(x) \leq g(x)$$

for all x in some open interval containing c , except possibly at $x = c$, and both limits exist, then

$$\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x).$$

Limits preserve the non-strict inequality $\leq$, but may not preserve the strict inequality $<$. For example, let

$$f(x) = x^2, \quad g(x) = 2x^2.$$

For $x \neq 0$, $f(x) < g(x)$, but

$$\lim_{x \rightarrow 0} x^2 = \lim_{x \rightarrow 0} 2x^2 = 0.$$

Lemma 1.1.5

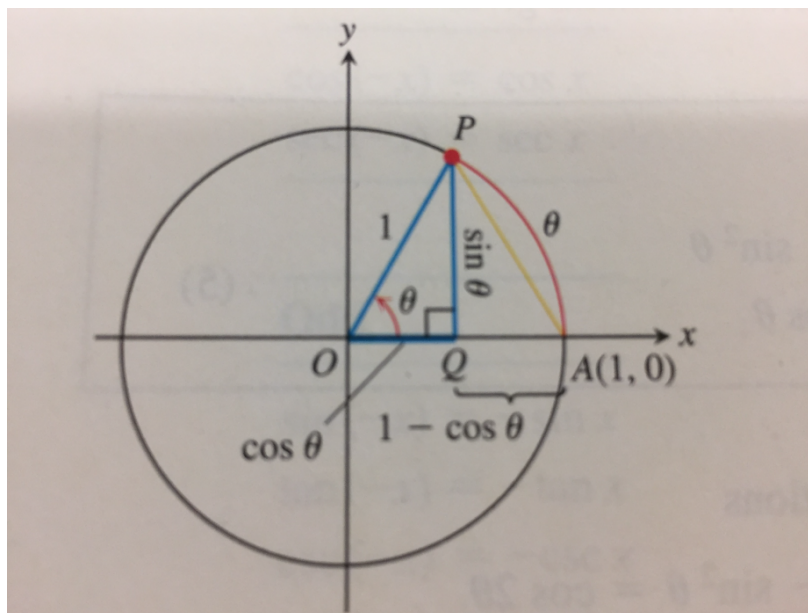
For θ measured in radians,

$$-|\theta| \leq \sin \theta \leq |\theta|, \quad -|\theta| \leq 1 - \cos \theta \leq |\theta|.$$

In particular, for $0 < \theta \leq \frac{\pi}{2}$,

$$0 < \sin \theta < \theta.$$

Proof. By the mean value theorem applied to $\sin$ between 0 and θ , $|\sin \theta| \leq |\theta|$. Also $1 - \cos \theta = 2 \sin^2(\theta/2) \leq |\theta|$ by the first inequality. The final inequality is the standard unit-circle comparison for $0 < \theta \leq \pi/2$.



One-sided Limits

Left-hand limit

$$\lim_{x \rightarrow c^-} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$c - \delta < x < c \implies |f(x) - L| < \varepsilon.$$

Right-hand limit

$$\lim_{x \rightarrow c^+} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$c < x < c + \delta \implies |f(x) - L| < \varepsilon.$$

The two-sided limit exists if and only if the one-sided limits both exist and are equal:

$$\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L.$$

Examples at $x \rightarrow 0$

Example 1: The 0-1 step function.

$$U(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^-} U(x) = 0, \quad \lim_{x \rightarrow 0^+} U(x) = 1,$$

so the two-sided limit does not exist.

Example 2: The function $1/x$.

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty, \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty.$$

Thus, $1/x$ has no finite two-sided limit at 0.

Example 3: The oscillating function $\sin(1/x)$.

$$\lim_{x \rightarrow 0^-} \sin \frac{1}{x} \quad \text{and} \quad \lim_{x \rightarrow 0^+} \sin \frac{1}{x}$$

both do not exist.

The density of $\mathbb{Q}$ in $\mathbb{R}$. We say that $\mathbb{Q}$ is dense in $\mathbb{R}$ if every non-empty open interval contains a rational number. For any $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that

$$a < q < b.$$

Equivalently, for every $x \in \mathbb{R}$ and every $\varepsilon > 0$, there exists $q \in \mathbb{Q}$ such that

$$|q - x| < \varepsilon.$$

The irrational numbers are also dense in $\mathbb{R}$.

Example 4: The Dirichlet function.

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Every interval to the left or right of 0 contains both rational and irrational numbers. Hence,

$$\lim_{x \rightarrow 0^-} D(x) \quad \text{and} \quad \lim_{x \rightarrow 0^+} D(x)$$

both do not exist.

Theorem 1.1.6

Suppose that $f(x)$ is defined on an open interval containing c , except possibly at c itself. Then $f(x)$ has a limit as x approaches c if and only if its left-hand and right-hand limits both exist and are equal:

$$\lim_{x \rightarrow c} f(x) = L \quad \Longleftrightarrow \quad \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L.$$

Proof

If $\lim_{x \rightarrow c} f(x) = L$, the ε - δ condition immediately implies the same estimate for $x < c$ and for $x > c$. Therefore both one-sided limits equal L .

Conversely, suppose both one-sided limits equal L . Given $\varepsilon > 0$, choose $\delta_1, \delta_2 > 0$ from the left- and right-hand definitions and let

$$\delta = \min\{\delta_1, \delta_2\}.$$

If $0 < |x - c| < \delta$, then either $c - \delta < x < c$ or $c < x < c + \delta$. In either case,

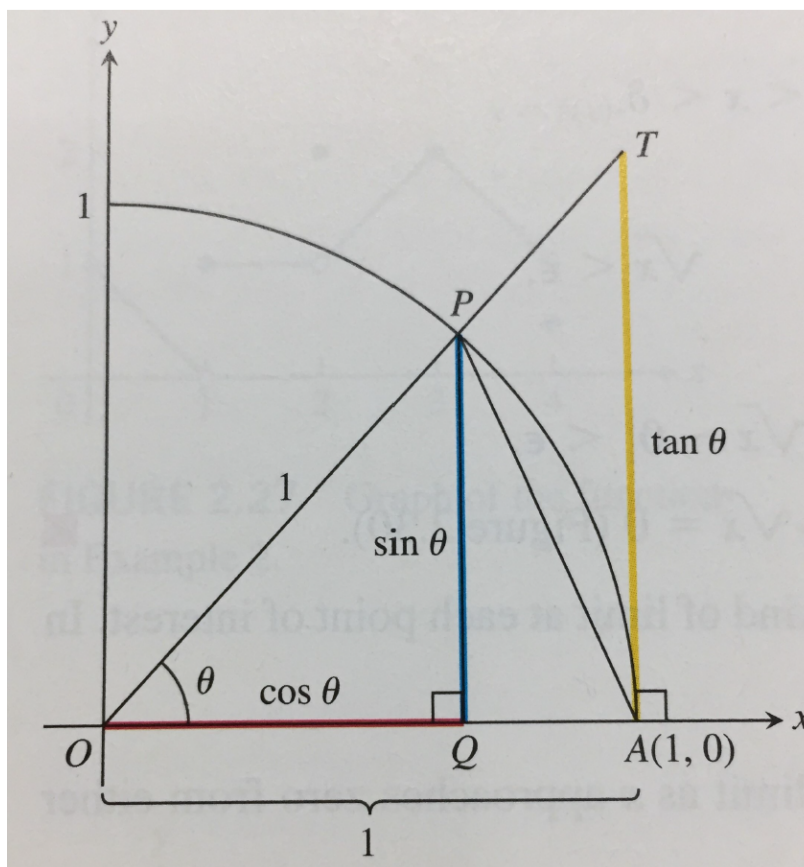
$$|f(x) - L| < \varepsilon.$$

Thus $\lim_{x \rightarrow c} f(x) = L$.

Theorem 1.1.7

For θ measured in radians,

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$



Proof

For $0 < \theta < \frac{\pi}{2}$,

$$\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1.$$

Since $\lim_{\theta \rightarrow 0} \cos \theta = 1$, the Sandwich Theorem gives

$$\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1.$$

For $\theta < 0$,

$$\frac{\sin \theta}{\theta} = \frac{\sin(-\theta)}{-\theta},$$

so the left-hand limit is also 1. Therefore,

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1}.$$

Consequences

For any constant $k \neq 0$,

$$\lim_{\theta \rightarrow 0} \frac{\sin(k\theta)}{\theta} = k \lim_{\theta \rightarrow 0} \frac{\sin(k\theta)}{k\theta} = k.$$

Also,

$$\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x} = 1,$$

and, for n nested sine functions,

$$\lim_{x \rightarrow 0} \frac{\sin(\sin(\cdots \sin x \cdots))}{x} = 1.$$

Proof of $\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{x} = 1$

Since $\lim_{x \rightarrow 0} \sin x = 0$, Theorem 1.1.7 gives

$$\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x} = 1.$$

Moreover,

$$\frac{\sin(\sin x)}{x} = \frac{\sin(\sin x)}{\sin x} \cdot \frac{\sin x}{x}.$$

Therefore,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{x} &= \left(\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x} \right) \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \\ &= 1 \cdot 1 = 1. \end{aligned}$$

Lecture 4

Example 1.1.19

$$f(x) = \begin{cases} x, & x < 0, \\ 1, & 0 \leq x < 1, \\ 2, & x = 1, \\ x, & x > 1. \end{cases}$$

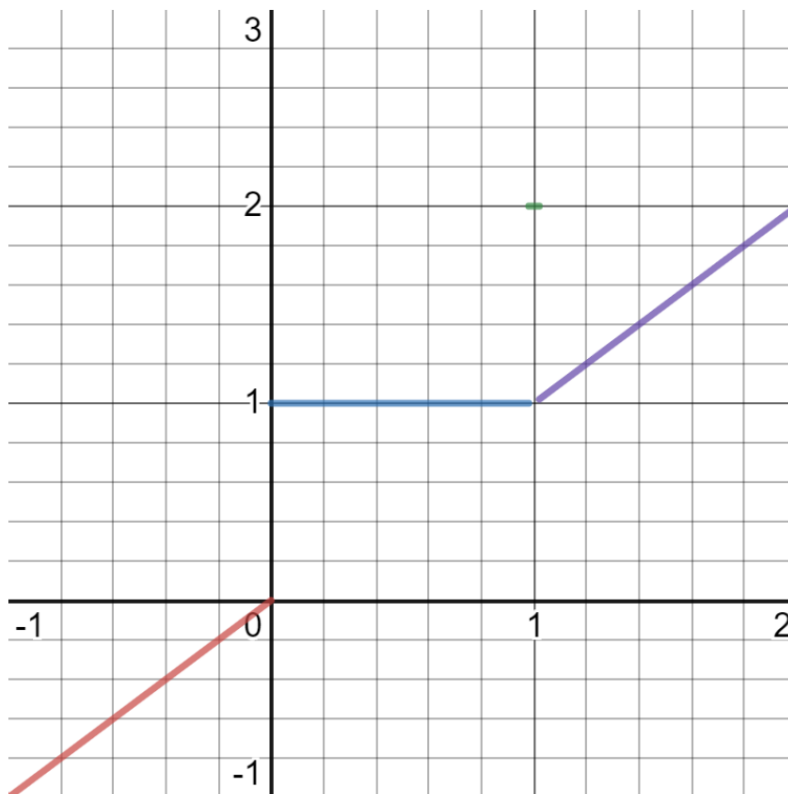


Figure: jump discontinuity

Definition 1.1.8 (Continuity at a Point)

Let c be a real number that is either an interior point or an endpoint of an interval in the domain of f .

(a) Interior point

f is continuous at c if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

(b) Right-continuity

f is right-continuous at c (or continuous from the right) if

$$\lim_{x \rightarrow c^+} f(x) = f(c).$$

(c) Left-continuity

f is left-continuous at c (or continuous from the left) if

$$\lim_{x \rightarrow c^-} f(x) = f(c).$$

Additional examples from the lectures

Uniqueness of the limit

Proposition. A function cannot have two different limits at the same point. If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} f(x) = K$, then $L = K$.

Proof. Suppose $L \neq K$ and choose $\varepsilon = |L - K|/3$. The two ε -neighborhoods of L and K are disjoint, but the definition of limit would force $f(x)$ to lie in both for x sufficiently close to c . This contradiction proves $L = K$.

A useful ε - δ example

To prove $\lim_{x \rightarrow 2} x^2 = 4$, observe

$$|x^2 - 4| = |x - 2||x + 2|.$$

If $|x - 2| < 1$, then $|x + 2| < 5$. Thus it is enough to take $\delta = \min\{1, \varepsilon/5\}$.

Power and root rules

The limit laws imply

$$\lim_{x \rightarrow c} [f(x)]^n = \left(\lim_{x \rightarrow c} f(x) \right)^n.$$

If $f(x) \geq 0$ near c and the right-hand limit is nonnegative, then

$$\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow c} f(x)}.$$

For example,

$$\lim_{x \rightarrow 4} \sqrt{x} = 2, \quad \lim_{x \rightarrow 1} \frac{x^3 + 2x}{x + 1} = \frac{3}{2}.$$

One-sided limit example

For $f(x) = |x|/x$,

$$\lim_{x \rightarrow 0^-} f(x) = -1, \quad \lim_{x \rightarrow 0^+} f(x) = 1,$$

so the two-sided limit does not exist. This is the simplest model of a jump discontinuity.

Continuity test and types of discontinuity

At an interior point c , continuity requires all three conditions:

1. $f(c)$ is defined;
2. $\lim_{x \rightarrow c} f(x)$ exists;
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

For the piecewise function in Example 1.1.19, the point 0 is a jump discontinuity because the one-sided limits are 0 and 1. At 1, the two-sided limit is 1 but $f(1) = 2$, so the discontinuity is removable: changing the value at 1 to 1 would make the function continuous there.

Riemann-type example

Define $R(x) = 1/q$ when $x = p/q \in \mathbb{Q} \cap (0, 1]$ is written in lowest terms, and $R(x) = 0$ when x is irrational. At every irrational point, $R(x) \rightarrow 0$; at a rational point p/q , nearby rationals with large denominators make the function values arbitrarily small, but the value at the point is positive. Thus R is continuous at irrational points and discontinuous at rational points.